Exam-like questions - Nuclear

1. These nuclides are fissile /1

   ◯ ^{233}U, ^{235}U, ^{238}U

   ◯ ^{233}U, ^{235}U, ^{239}Pu

   ◯ ^{238}U, ^{239}Pu, ^{232}Th

   ◯ ^{233}U, ^{238}U, ^{232}Th

   ◯ ^{235}U, ^{238}U, ^{239}Pu

2. Nuclear fusion is based on the interactions of (D: deuterium, T: tritium, U: uranium, Th: Thorium, n: neutron) /1

   ◯ D+T

   ◯ D+U

   ◯ T+U

   ◯ U+n

   ◯ Th+n

3. Arrange the most common types of radioactive radiation in order of increasing hardness. /1

   ◯ alpha < beta < gamma

   ◯ gamma < beta <alpha

   ◯ beta <gamma < alpha

   ◯ alpha < gamma = beta

   ◯ alpha = beta < gamma

4. What is the mass defect of an atom? /1

- ◯ The mass that is lost when a nucleus is split
- ◯ The mass that is gained when a nucleus is split
- ◯ The difference between the mass of the atom and the sum of that mas of all the protons, neutrons and electrons it constitutes. The mass of the atom is smaller than that of the constituents due to the binding energy
- ◯ The difference between the mass of the atom and the sum of that mas of all the protons, neutrons and electrons it constitutes. The mass of the atom is larger than that of the constituents due to the binding energy
- ◯ The difference in mass of protons and neutrons

5. Which types of fission reactors the most common ones? /1

- ◯ The pressurized water reactor and the molten salt reactor
- ◯ The pressurized water reactor and the boiling water reactor
- ◯ The molten salt reactor and the boiling water reactor
- ◯ The sodium-cooled fast reactor and the pressurized water reactor
- ◯ The pressurized water reactor and the gas-cooled fast reactor

6. How many protons neutrons and electron are assembled in an atom of the isotope $^{235}_{92}U$ /1

- ◯ 235 neutrons, 92 protons and 92 electrons
- ◯ 92 neutrons, 235 protons and 235 electrons
- ◯ 51 neutrons, 92 protons and 92 electrons
- ◯ 143 neutrons, 92 protons and 92 electrons
- ◯ 235 neutrons, 92 protons and 235 electrons

7. Calculate the mass defect of $^{232}_{90}Th$. It has the atomic mass of 232.038 amu (atomic mass unit). Find the mass of the elemental particles (protons, neutrons and electrons) in the list of formulas. /1

- ◯ 0.93 amu
- ◯ 1.28 amu
- ◯ 1.89 amu
- ◯ 2.37 amu
- ◯ 3.34 amu